Distributed Learning for Resilient LEO Satellite Access under Adversarial Jamming

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***Abstract*—Low-Earth-orbit (LEO) mega-constellations are expected to support ubiquitous connectivity, yet resilient access remains challenging under satellite mobility, adversarial jamming, and limited onboard resources. This paper investigates user association and scheduling in LEO networks under adversarial jamming, while accounting for link activation cost, finite radio frequency chains, and long-term queue stability requirements. We propose a decentralized online learning framework, termed Lyapunov-guided variance-aware multi-armed bandit (LV-MAB). Specifically, Lyapunov drift analysis converts the long-term queue stability requirement into a per-frame queue-weighted surrogate problem, which enables distributed user-side association and satellite-side scheduling. For user association, LV-MAB uses variance-aware upper confidence bound with link-level sliding windows and resets to track non-stationary rate statistics induced by mobility and dynamic jamming. For satellite-side scheduling, combinatorial Thompson sampling with posterior inflation is adopted to resolve contention among requesting users under radio frequency chain constraints. We establish dynamic regret bounds for LV-MAB and prove system queue stability. Extensive simulations show that LV-MAB achieves reliable throughput, backlog control, fairness improvement, and robustness under dynamic adversarial jamming.**

***Index Terms*—LEO mega-constellations, anti-jamming, multi-armed bandit, agentic AI, Lyapunov optimization.**

I. Introduction

Sixth-generation (6G) networks are expected to evolve toward AI-native and autonomous networking, where communication nodes perceive local states, learn from interactions, and adapt their control decisions online. As a key component of 6G, non-terrestrial networks, especially low-Earth-orbit (LEO) mega-constellations, are envisioned to provide ubiquitous global connectivity [1]–[3]. In practical LEO access environments, however, service continuity is challenged not only by satellite mobility but also by adversarial jamming. The wide coverage footprint and broadcast nature of satellite access expose selected satellites to directional interference, while intelligent jammers may retarget their beams to disrupt different satellites over time [4].

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Against this background, maintaining continuous service in LEO access networks requires coordinated user association and satellite-side scheduling. User association determines which satellite a user should access, while satellite-side scheduling allocates limited onboard radio resources among competing users. This coordination is challenging because satellite motion continuously reshapes feasible user-satellite links, and jammer target hopping creates abrupt service degradation on the targeted satellite links. In addition, handover to a new satellite or service reactivation after an idle frame requires beam alignment and synchronization, reducing the effective payload [5]. These coupled effects must be handled under strict onboard size, weight, and power constraints and finite radio frequency (RF) chains [6]. The resulting problem is to make association and scheduling decisions that balance service quality, link activation cost, resource contention, and queue stability when only limited local information is available.

However, current approaches have difficulty meeting these requirements in distributed LEO mega-constellations. Traditional optimization methods rely heavily on real-time global channel state information (CSI) [7], [8]. In large constellations, collecting such information imposes prohibitive signaling overhead. Moreover, propagation delays, satellite motion, and jammer retargeting may make the collected CSI outdated before the corresponding access and scheduling decisions are executed, leading to severe performance degradation.

These limitations motivate an agentic online learning view of resilient LEO access, where distributed nodes adapt their association and scheduling decisions from local observations and interaction outcomes under mobility, adversarial interference, and onboard resource constraints. Deep reinforcement learning (DRL) approaches can learn adaptive policies, but their training cost, model storage, and inference latency strain the limited onboard processing budget, thereby limiting real-time adaptability in highly dynamic LEO access scenarios [9]. Compared with DRL, multi-armed bandit (MAB) algorithms provide a natural lightweight alternative because association and scheduling decisions must be made before the link rates are observed, and rate samples are revealed only for the executed links. MAB focuses on the immediate exploration-exploitation trade-off and can adapt from local observations with low computational overhead.

However, directly applying standard MAB algorithms such as upper confidence bound (UCB) and Thompson sampling is insufficient in this setting. Their stationary reward assumptions [10], [11] make them slow to react when jamming or visibility changes alter the local service distribution [12]. They also do not account for queue backlogs, which may lead

to unbounded backlog growth and instability [13], [14]. In addition, a straightforward MAB formulation that treats each feasible association and scheduling pair as one arm would create a combinatorial explosion in the arm space [15].

To tackle these challenges, we propose Lyapunov-guided variance-aware MAB (LV-MAB), a decentralized framework for queue-stable and jammer-resilient LEO access. Specifically, LV-MAB applies Lyapunov drift analysis to convert the long-term queue stability requirement into a per-frame queue-weighted surrogate objective. The resulting objective incorporates backlog pressure, uncertain service rate, and link activation cost, and provides the basis for factorizing the joint online decision into user-side association over visible satellites and satellite-side scheduling among requesting users. This factorization enables lightweight distributed online learning under jamming-induced non-stationarity, while avoiding excessive signaling overhead and centralized combinatorial search over coupled association and scheduling decisions. The main contributions of this paper are summarized as follows.

- *Per-Frame Access Reformulation:* We formulate joint user association and scheduling for LEO access with adversarial jamming, link activation cost, finite RF chains, and stochastic queue dynamics. By Lyapunov drift analysis, the long-term queue stability constraint is transformed into per-frame queue-weighted service objectives enabling online learning.
- *Distributed Agentic LV-MAB Design:* We develop a LV-MAB framework that decomposes the joint online decision into user-side association and satellite-side scheduling. Specifically, variance-aware resets refresh stale link statistics for non-stationary association learning, while combinatorial Thompson sampling (CTS) resolves scheduling contention among requesting users.
- *Theoretical Guarantees and Simulation Validation:* We establish explicit dynamic regret bounds for LV-MAB and further prove queue stability. Extensive simulations under dynamic jamming show that LV-MAB consistently improves throughput, backlog control, fairness, and robustness over benchmark schemes.

The remainder of this paper is organized as follows. Related work is reviewed in Section II, followed by the system model and problem formulation in Section III. Section IV presents LV-MAB, and Section V analyzes its regret and queue stability. Simulation results are reported in Section VI, and Section VII concludes the paper.

II. Related Work

Existing approaches related to resilient LEO access can be grouped into three lines: model based resource management, deep agentic learning, and lightweight bandit learning.

A. Model Based Resource Management

Model based methods usually rely on explicit system state, static optimization, or strategic game formulations. In these frameworks, agents are often modeled as rational decision makers aiming to reach a stable equilibrium under quasi-static assumptions.

For high-throughput satellite and non-terrestrial network resource management, Roumeliotis *et al.* [8] formulated gateway-beam association as a many-to-one matching problem with channel-aware preferences, while Honnaiah *et al.* [16] used clustering optimization to adapt beam coverage to spatial traffic heterogeneity. For distributed massive access, Choi [17] applied evolutionary game theory to characterize local agent interactions and stable access behavior. In adversarial settings, Jia *et al.* [4] used a Stackelberg game to analyze intelligent jamming and anti-jamming strategies.

These approaches provide useful structural insights and theoretical benchmarks. However, their dependence on model accuracy, centralized state knowledge, or iterative convergence makes them difficult to synchronize with rapidly evolving satellite geometry and jammer retargeting in distributed LEO access.

B. Deep Agentic Learning

DRL and generative artificial intelligence provide powerful agentic learning tools because they can infer control policies directly from high-dimensional observations and reduce the need for explicit analytical models.

Within deep agentic learning, Lee *et al.* [9] adopted multi-agent DRL under centralized training and decentralized execution for cooperative resource control, and Dalili Shoaei *et al.* [18] designed a Q-learning based medium access control (MAC) architecture for heterogeneous quality of service (QoS) requirements. For space-air-ground integrated network (SAGIN) service function chain scheduling, Jia *et al.* [19] used DRL to handle dynamic scheduling across heterogeneous space-air-ground resources. For adversarial LEO networking, Silvirianti *et al.* [20] studied DRL-based anti-jamming routing, while Li *et al.* [21] developed distributed multi-agent DRL for anti-jamming frequency-power control in resource-constrained mega-constellations. More recently, Lian *et al.* [22] integrated large language models (LLMs) with MAB to exploit prior knowledge for interference-aware decision making.

However, despite their strong representation and coordination capabilities, deep agentic frameworks still face non-negligible deployment barriers in highly dynamic LEO access scenarios. Deep neural network policies typically require substantial training, model storage, and inference resources, while their adaptation latency may be problematic for fast LEO access. Moreover, existing DRL studies mainly focus on routing, service function chain scheduling, or spectrum-power control, rather than on queue-stable distributed user association and RF chain constrained satellite scheduling under online jamming dynamics.

C. Lightweight Bandit Learning

Against this background, bandit online learning offers a lightweight approach for local decision making in LEO access networks.

Standard MAB algorithms rely on the stationarity assumption, which rarely holds in dynamic LEO environments. To track non-stationary statistics, Garivier and Moulines [12]

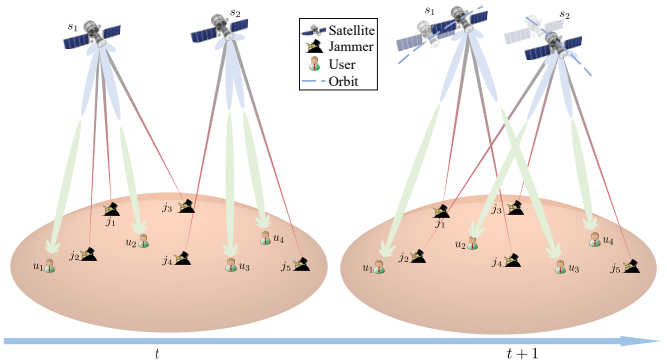


Fig. 1. Decentralized agentic networking in LEO mega-constellations. Satellites traverse orbital trajectories between frames t and $t+1$ while maintaining associations with users under distributed adversarial jamming.

and Raj and Kalyani [23] introduced sliding-window and discounted UCB strategies, respectively, employing forgetting mechanisms to discard stale observations. Audibert *et al.* [24] proposed UCB-V, which incorporates variance estimates into the exploration term. This use of second-order statistics is useful for calibrating uncertainty when fading, mobility, and jamming cause heterogeneous rate variations [4].

The joint allocation of RF chains and users induces a combinatorial explosion in the action space. To mitigate this dimensionality, Song *et al.* [15] formulated combinatorial MAB frameworks that exploit linear dependencies to efficiently isolate optimal beam subsets. For beam alignment, Zhang and Heath [25] and Sneh *et al.* [26] capitalized on the unimodal structure of beam patterns to prune the search space, achieving logarithmic regret without exhaustive scanning.

Practical implementation also demands strict adherence to physical constraints beyond mere reward maximization. Xu *et al.* [27] utilized budgeted MAB frameworks to optimize throughput under finite energy budgets. Addressing queue stability, Gao *et al.* [14] embedded Lyapunov optimization into the UCB algorithm.

Nevertheless, these studies mainly focus on benign time-varying environments, energy-aware access, beam alignment, or task offloading scenarios. They do not provide a unified learning formulation that jointly supports adaptation to adversarial jamming, distributed user association, RF chain constrained satellite-side scheduling, variance-triggered belief reset, and long-term queue stability. Motivated by this gap, we develop a Lyapunov-guided variance-aware combinatorial learning framework for queue-stable and jammer-resilient access in LEO satellite networks.

III. System Model and Problem Formulation

We consider a distributed LEO access network consisting of a set of ground users modeled as user agents (UAs) $\mathcal{U}$, a set of satellites modeled as satellite agents (SAs) $\mathcal{S}$, and a set of adversarial jammers $\mathcal{J}$, as illustrated in Fig. 1. The system evolves over T decision frames indexed by $t \in \{1, 2, \ldots, T\}$, each with duration τ_{frame}. In each frame, UA association and SA scheduling are determined, and the resulting decisions remain fixed throughout the frame. To capture physical-layer service variations within a frame, each frame is further divided into M micro-slots of duration τ_{slot}, where $\tau_{\text{frame}} = M\tau_{\text{slot}}$. Packet transmission, channel fading, interference, and delivered service are then modeled over the micro-slots.

At the beginning of frame t, the visibility relations between ground nodes and SAs are determined according to the minimum elevation angle threshold $\phi_{\min}$. The set of SAs visible to UA $u \in \mathcal{U}$ is defined as $\mathcal{S}_u(t) = \{s \in \mathcal{S} \mid \phi_{us}(t) \geq \phi_{\min}\}$. From the perspective of SA $s \in \mathcal{S}$, the visible candidate-UA set and visible-jammer set are defined as $\mathcal{U}_s(t) = \{u \in \mathcal{U} \mid \phi_{us}(t) \geq \phi_{\min}\}$ and $\mathcal{J}_s(t) = \{j \in \mathcal{J} \mid \phi_{js}(t) \geq \phi_{\min}\}$, respectively. The term $\phi_{ks}(t)$ denotes the elevation angle between ground node $k \in \mathcal{U} \cup \mathcal{J}$ and SA s.

Let $\mathbf{A}(t)$, $\mathbf{B}(t)$, and $\mathbf{M}(t)$ denote the binary matrices for UA association, jammer targeting, and SA scheduling, respectively. Specifically, $\alpha_{us}(t) = 1$ means that UA u is associated with SA s, $\beta_{js}(t) = 1$ means that jammer j directs its jamming beam toward SA s, and $\mu_{us}(t) = 1$ means that SA s admits UA u for data transmission in frame t. The jammer-targeting matrix $\mathbf{B}(t)$ is an exogenous adversarial process rather than a network control variable, and it determines which visible jammers contribute interference to each SA.

They are constrained as follows. Each UA can be associated with at most one visible SA in a frame, i.e., $\sum_{s \in \mathcal{S}_u(t)} \alpha_{us}(t) \leq 1$ for all $u \in \mathcal{U}$. Each jammer can direct its beam toward at most one SA in a frame, i.e., $\sum_{s \in \mathcal{S}} \beta_{js}(t) \leq 1$ for all $j \in \mathcal{J}$, and it cannot target an SA outside its visibility set, i.e., $\beta_{js}(t) = 0$ for $j \notin \mathcal{J}_s(t)$. On the SA side, each SA can serve at most $N_{\max}$ UAs simultaneously due to its finite RF chain capacity, i.e., $\sum_{u \in \mathcal{U}} \mu_{us}(t) \leq N_{\max}$ for all $s \in \mathcal{S}$.

A. Uplink Transmission Model

All UAs, SAs, and jammers are equipped with uniform planar arrays (UPAs). To reduce signaling overhead, each ground node employs open-loop transmit beamforming $\mathbf{v}_k(t, m)$ by steering toward the line-of-sight (LoS) direction obtained from satellite ephemeris. The uplink channel matrix $\mathbf{H}_{ks}(t, m)$ from ground node $k \in \mathcal{U} \cup \mathcal{J}$ to SA s captures the micro-slot channel state, including deterministic LoS evolution and stochastic multipath scattering. After transmit beamforming, the effective channel vector is

$$\mathbf{h}_{ks}(t, m) = \mathbf{H}_{ks}(t, m)\mathbf{v}_k(t, m), \tag{1}$$

where $\mathbf{h}_{ks}(t, m) \in \mathbb{C}^{N_s \times 1}$ and N_s denotes the number of antenna elements at each SA.

Each SA uses a hybrid receive beamforming architecture with N_{RF} RF chains connected to an N_s-element UPA. Following [28], the number of simultaneously scheduled UAs is limited to $N_{\max} = \lfloor N_{\text{RF}}/2 \rfloor$. This setting leaves sufficient spatial degrees of freedom for low-complexity digital combining while respecting the onboard RF chain limitation.

To recover uplink signals, each SA applies a two-stage hybrid combining structure. The analog stage aligns the receive beams with the LoS directions of scheduled UAs, while the digital stage applies a linear minimum mean square error (MMSE) detector to suppress residual multi-user interference. The MMSE weights are constructed using only local

CSI of scheduled UAs, and jammer channels are treated as unobservable interference. The final receive combining vector for detecting UA u at SA s is

$$\mathbf{w}_{us}(t,m) = \mathbf{W}_s^{\mathrm{RF}}(t,m)\mathbf{w}_{us}^{\mathrm{BB}}(t,m), \tag{2}$$

where $\mathbf{W}_s^{\mathrm{RF}}(t,m)$ and $\mathbf{w}_{us}^{\mathrm{BB}}(t,m)$ denote the analog combining matrix and the digital baseband MMSE weight vector, respectively, and $\mathbf{w}_{us}(t,m) \in \mathbb{C}^{N_s \times 1}$.

For a scheduled link (u,s), the detected signal after hybrid combining in micro-slot m of frame t is

$$\hat{x}_u(t,m) = \mathbf{w}_{us}^H(t,m)\Bigg[\sum_{k\in\mathcal{U}_s(t)} \mu_{ks}(t)\sqrt{p_k}\mathbf{h}_{ks}(t,m)x_k(t,m) + \sum_{j\in\mathcal{J}_s(t)} \beta_{js}(t)\sqrt{p_j}\mathbf{h}_{js}(t,m)x_j(t,m) + \mathbf{n}(t,m)\Bigg], \tag{3}$$

where the first summation includes the desired signal from UA u and the interference from other scheduled UAs, p_k and p_j denote the transmit powers of UA k and jammer j, respectively, and $\mathbf{n}(t,m) \sim \mathcal{CN}(\mathbf{0}, \sigma^2\mathbf{I}_{N_s})$ is the antenna noise vector. Note that the factors $\mu_{ks}(t)$ and $\beta_{js}(t)$ restrict the received signal to scheduled UAs and jammers targeting SA s, respectively.

The corresponding signal-to-interference-plus-noise ratio (SINR) of UA u at SA s is

$$\gamma_{us}(t,m) = \frac{p_u \left|\mathbf{w}_{us}^H(t,m)\mathbf{h}_{us}(t,m)\right|^2}{I_{us}^{\mathrm{jam}}(t,m) + I_{us}^{\mathrm{mui}}(t,m) + \sigma^2 \left\|\mathbf{w}_{us}(t,m)\right\|^2}, \tag{4}$$

where $I_{us}^{\mathrm{jam}}(t,m) = \sum_{j\in\mathcal{J}_s(t)} \beta_{js}(t)p_j \left|\mathbf{w}_{us}^H(t,m)\mathbf{h}_{js}(t,m)\right|^2$ and $I_{us}^{\mathrm{mui}}(t,m) = \sum_{k\in\mathcal{U}_s(t),\, k\neq u} \mu_{ks}(t)p_k \left|\mathbf{w}_{us}^H(t,m)\mathbf{h}_{ks}(t,m)\right|^2$ denote the jamming interference and multi-user interference powers after receive combining, respectively. Then, the instantaneous achievable rate of link (u,s) is

$$R_{us}(t,m) = B_{\mathrm{w}} \log_2(1 + \gamma_{us}(t,m)), \tag{5}$$

where B_{w} denotes the transmission bandwidth.

B. Service Model

Based on the micro-slot achievable rate $R_{us}(t,m)$ derived above, we next model the delivered service and the resulting queue evolution over each decision frame. As defined by the frame-level control timescale, the association matrix $\mathbf{A}(t)$ and the scheduling matrix $\mathbf{M}(t)$ remain fixed over the M micro-slots of frame t. A link that was already scheduled in the previous frame can transmit payload data from the first micro-slot of the current frame. For a newly activated serving link, which may result from satellite handover or service reactivation after an idle frame, beam alignment and synchronization occupy the first $C_{\mathrm{act}} \in \{0,1,\ldots,M-1\}$ micro-slots before payload transmission starts. We refer to this overhead as the link activation cost.

Since UA u can transmit data through SA s only when it is both associated with and scheduled by that SA, the service allocated to UA u in micro-slot m of frame t is

$$D_u(t,m) = \sum_{s\in\mathcal{S}_u(t)} \alpha_{us}(t)\mu_{us}(t)\chi_{us}(t,m)R_{us}(t,m)\tau_{\mathrm{slot}}, \tag{6}$$

where $\chi_{us}(t,m) \in \{0,1\}$ indicates whether link (u,s) can transmit payload data in micro-slot m. Specifically,

$$\chi_{us}(t,m) = \begin{cases} 1, & \text{if } \mu_{us}(t-1) = 1, \\ \mathbb{I}(m > C_{\mathrm{act}}), & \text{if } \mu_{us}(t-1) = 0. \end{cases} \tag{7}$$

The first case corresponds to a link that was already active in the previous frame, while the second case corresponds to a newly activated scheduled link. For initialization, we set $\mu_{us}(0) = 0$ for all u and s, meaning that all scheduled links in the first frame are treated as newly activated.

To capture stochastic traffic demand and service continuity, each UA maintains a data queue. Let $Q_u(t,m)$ denote the backlog of UA u at the beginning of micro-slot m in frame t, and let $Q_u(t) \triangleq Q_u(t,1)$ denote the frame-level backlog observed at the beginning of frame t. Let $Y_u(t,m)$ denote the data arrival of UA u in micro-slot m. Given the delivered service $D_u(t,m)$, the queue evolves as

$$Q_u(t,m+1) = \max\{Q_u(t,m) - D_u(t,m), 0\} + Y_u(t,m). \tag{8}$$

Recall that access and scheduling decisions are made at the frame level. Accordingly, the micro-slot service and arrivals are aggregated over each decision frame as

$$D_u(t) = \sum_{m=1}^{M} D_u(t,m), \quad Y_u(t) = \sum_{m=1}^{M} Y_u(t,m). \tag{9}$$

For each link (u,s), the frame-average achievable rate is defined as

$$R_{us}(t) = \frac{1}{M}\sum_{m=1}^{M} R_{us}(t,m). \tag{10}$$

C. Problem Formulation

The goal of the system is to maximize the long-term average delivered service by jointly optimizing the controllable UA association matrix $\mathbf{A}(t)$ and SA scheduling matrix $\mathbf{M}(t)$, subject to visibility constraints, RF chain capacity limits, association-scheduling consistency, and long-term queue stability. The resulting stochastic optimization problem is formulated as

$$\max_{\{\mathbf{A}(t),\mathbf{M}(t)\}} \quad \frac{1}{T}\sum_{t=1}^{T}\sum_{u\in\mathcal{U}} \mathbb{E}\left[D_u(t)\right] \tag{11a}$$

$$\text{s.t.} \quad \alpha_{us}(t) \in \{0,1\}, \quad \forall u \in \mathcal{U}, s \in \mathcal{S}_u(t), \tag{11b}$$

$$\mu_{us}(t) \in \{0,1\}, \quad \forall u \in \mathcal{U}, s \in \mathcal{S}_u(t), \tag{11c}$$

$$\alpha_{us}(t) = 0, \quad \forall u \in \mathcal{U}, s \notin \mathcal{S}_u(t), \tag{11d}$$

$$\mu_{us}(t) = 0, \quad \forall u \in \mathcal{U}, s \notin \mathcal{S}_u(t), \tag{11e}$$

$$\sum_{s\in\mathcal{S}} \alpha_{us}(t) \le 1, \quad \forall u \in \mathcal{U}, \tag{11f}$$

$$\sum_{u\in\mathcal{U}} \mu_{us}(t) \le N_{\max}, \quad \forall s \in \mathcal{S}, \tag{11g}$$

$$\mu_{us}(t) \le \alpha_{us}(t), \quad \forall u \in \mathcal{U}, s \in \mathcal{S}, \tag{11h}$$

$$\limsup_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T}\sum_{u\in\mathcal{U}} \mathbb{E}[Q_u(t)] < \infty. \tag{11i}$$

where constraints (11b) $\sim$ (11e) impose binary association and scheduling decisions within the visibility region, (11f) ensures

that each UA is associated with at most one SA in a frame, (11g) captures the finite RF chain capacity of each SA, (11h) ensures that a UA can be scheduled by an SA only after being associated with that SA, and (11i) imposes long-term queue stability. Note that the jammer-targeting matrix $\mathbf{B}(t)$ is not optimized in (11). It acts as an unknown adversarial state that affects the objective through the jamming interference and the resulting service process.

Problem (11) is a sequential mixed-integer nonlinear stochastic optimization problem. The integer variables $\alpha_{us}(t)$ and $\mu_{us}(t)$ couple UA association with SA scheduling, while the service term $D_u(t)$ depends nonlinearly on hybrid receive combining, multi-user interference, and adversarial jamming. The problem is also temporally coupled because current scheduling decisions determine whether each link is newly activated in the next frame and affect the future queue backlog. Moreover, the instantaneous rate $R_{us}(t,m)$ is not known before the access and scheduling decisions are made, and collecting global CSI and queue states would incur prohibitive signaling overhead in a rapidly moving LEO network. These factors make offline centralized optimization unsuitable and motivate the distributed online learning framework developed in the next section.

IV. LV-MAB Framework for Resilient LEO Access

In this section, we develop the LV-MAB framework to solve Problem (11) in a distributed and online manner. We first apply Lyapunov drift analysis to transform the long-term queue stability problem into a per-frame service maximization problem. To handle the resulting utility that contains the unknown rate term $R_{us}(t)$, which is unavailable before access and scheduling decisions and can only be observed after the decisions are executed, we formulate two local MAB learning problems, namely UA-side visible-SA selection and SA-side scheduling among requesting UAs under RF chain constraints. The following subsections derive the per-frame surrogate problem and present the resulting LV-MAB algorithm.

A. Temporal Decoupling via Lyapunov Drift

We first employ Lyapunov drift analysis to transform the long-term stability requirement in (11i) into a per-frame surrogate objective.

Let $\mathbf{Q}(t) = [Q_1(t), \ldots, Q_{|\mathcal{U}|}(t)]$ denote the frame-level queue state. We use the quadratic Lyapunov function $L(\mathbf{Q}(t)) = \frac{1}{2}\sum_{u\in\mathcal{U}} Q_u^2(t)$ as a scalar measure of aggregate queue congestion. The one-step conditional drift $\Delta(\mathbf{Q}(t)) = \mathbb{E}[L(\mathbf{Q}(t+1)) - L(\mathbf{Q}(t)) \mid \mathbf{Q}(t)]$ measures the expected change of this congestion measure over one frame. According to the Lyapunov drift criterion for queueing systems [13], a drift inequality of the form $\Delta(\mathbf{Q}(t)) \le \Phi - \epsilon\sum_{u\in\mathcal{U}} Q_u(t)$, with $\Phi < \infty$ and $\epsilon > 0$, provides a sufficient condition for strong queue stability. Motivated by this criterion, we derive an upper bound on the one-step drift in the following.

Specifically, from the queue dynamics in (8) and the frame-level aggregation in (9), we have $Q_u(t+1) \le \max\{Q_u(t) - D_u(t), 0\} + Y_u(t)$. Squaring this bound, using $(\max\{Q - D, 0\} + Y)^2 \le Q^2 + D^2 + Y^2 - 2Q(D - Y)$, summing over all UAs, and taking the conditional expectation given $\mathbf{Q}(t)$, we obtain

$$\Delta(\mathbf{Q}(t)) \le \Phi - \sum_{u\in\mathcal{U}} \mathbb{E}\left[Q_u(t)\big(D_u(t) - Y_u(t)\big) \mid \mathbf{Q}(t)\right], \quad (12)$$

where Φ is a finite constant satisfying $\Phi \ge \frac{1}{2}\sum_{u\in\mathcal{U}} \mathbb{E}\big[D_u^2(t) + Y_u^2(t) \mid \mathbf{Q}(t)\big]$, which upper bounds the second-order terms of the frame-level arrivals and delivered service. In (12), only the delivered service $D_u(t)$ depends on the association and scheduling decisions. Thus, minimizing the drift upper bound reduces to maximizing the queue-weighted delivered service in the current frame, which gives

$$\max_{(\mathbf{A}(t),\mathbf{M}(t))\in\mathcal{F}(t)} \mathbb{E}\left[\sum_{u\in\mathcal{U}} Q_u(t) D_u(t) \mid \mathbf{Q}(t)\right], \quad (13)$$

where $\mathcal{F}(t)$ denotes the feasible set of $(\mathbf{A}(t), \mathbf{M}(t))$ satisfying (11b) $\sim$ (11h) under the visibility and RF chain constraints of frame t.

We next express $D_u(t)$ in terms of the association and scheduling decisions. From the definition of $\chi_{us}(t,m)$, the number of micro-slots available for payload transmission over link (u,s) in frame t is

$$\sum_{m=1}^{M} \chi_{us}(t,m) = \begin{cases} M, & \mu_{us}(t-1) = 1, \\ M - C_{\mathrm{act}}, & \mu_{us}(t-1) = 0, \end{cases} = M\left[1 - \eta_{\mathrm{act}}\left(1 - \mu_{us}(t-1)\right)\right], \quad (14)$$

where $\eta_{\mathrm{act}} \triangleq C_{\mathrm{act}}/M$ denotes the fraction of a decision frame unavailable for payload transmission due to link activation when a serving link is newly activated. In the simulation setting, $C_{\mathrm{act}} = 2$ and $M = 50$, giving $\eta_{\mathrm{act}} = 4\%$.

Based on (6), the frame-level service can be written as

$$D_u(t) = \sum_{s\in\mathcal{S}_u(t)} \alpha_{us}(t)\mu_{us}(t)\tau_{\mathrm{slot}} \sum_{m=1}^{M} \chi_{us}(t,m) R_{us}(t,m) \approx \tau_{\mathrm{slot}} \sum_{s\in\mathcal{S}_u(t)} \alpha_{us}(t)\mu_{us}(t) R_{us}(t) \sum_{m=1}^{M} \chi_{us}(t,m), \quad (15)$$

where the last step replaces the average rate over the micro-slots available for payload transmission by the full-frame average rate $R_{us}(t)$. This approximation is exact when the achievable rate is constant within a frame and remains reasonable when the rate variation over one short decision frame is moderate. Its impact on the frame-level service is proportional to η_{act} and the within-frame rate variation.

Substituting (14) into (15) and then using the resulting frame-level service expression in (13), we obtain the following per-frame surrogate problem after omitting the common positive factor τ_{frame}

$$\max_{(\mathbf{A}(t),\mathbf{M}(t))\in\mathcal{F}(t)} \sum_{u\in\mathcal{U}} \sum_{s\in\mathcal{S}_u(t)} \alpha_{us}(t)\mu_{us}(t)\Psi_{us}(t), \quad (16)$$

where $\Psi_{us}(t) \triangleq Q_u(t) R_{us}(t)\left[1 - \eta_{\mathrm{act}}\left(1 - \mu_{us}(t-1)\right)\right]$ is the link-level contribution to the queue-weighted delivered service in frame t. Thus, the long-term queue stability requirement is handled through a per-frame surrogate problem, where

$\Psi_{us}(t)$ jointly captures the backlog weight, the frame-average achievable rate, and the link activation cost of link (u, s).

B. Distributed MAB Reformulation

Although the Lyapunov drift reformulation decouples the long-term problem across frames and yields the per-frame surrogate problem in (16), the resulting problem still cannot be solved directly. A deterministic centralized solution would require complete knowledge of the coefficients $\Psi_{us}(t)$ for all feasible links between UAs and SAs, together with joint optimization of $\mathbf{A}(t)$ and $\mathbf{M}(t)$. Since $\Psi_{us}(t)$ depends on the queue backlog $Q_u(t)$ and the frame-level rate $R_{us}(t)$, this would require network-wide queue information and link-rate information. Moreover, $R_{us}(t)$ is unknown before association and scheduling decisions are executed, and it can only be observed after the corresponding access and service events. Thus, (16) has the form of a bandit-feedback problem, where decisions are made before the link rates are observed and rate samples are revealed only for the executed links.

To make the online objective explicit, we define the surrogate objective for a feasible decision $(\mathbf{A}, \mathbf{M}) \in \mathcal{F}(t)$ as

$$f_t(\mathbf{A}, \mathbf{M}) = \sum_{u \in \mathcal{U}} \sum_{s \in \mathcal{S}_u(t)} \alpha_{us} \mu_{us} \Psi_{us}(t). \tag{17}$$

If $\Psi_{us}(t)$ were known for all feasible links between UAs and SAs, the full-information oracle decision in frame t would be

$$(\mathbf{A}^\star(t), \mathbf{M}^\star(t)) \in \arg \max_{(\mathbf{A},\mathbf{M}) \in \mathcal{F}(t)} f_t(\mathbf{A}, \mathbf{M}). \tag{18}$$

Because f_t contains the previous scheduling state through the payload transmission factor, the oracle and the online decision are compared under the same previous scheduling state. For any distributed online policy π that generates decisions $(\mathbf{A}^\pi(t), \mathbf{M}^\pi(t))$, its queue-weighted system regret relative to the oracle is

$$\mathcal{R}_{\text{sys}}^{Q,\pi}(T) = \mathbb{E}\left[\sum_{t=1}^{T} \left(f_t(\mathbf{A}^\star(t), \mathbf{M}^\star(t)) - f_t(\mathbf{A}^\pi(t), \mathbf{M}^\pi(t))\right)\right]. \tag{19}$$

Since f_t is obtained from the queue-weighted delivered service after omitting the common positive factor τ_{frame}, (19) measures the same oracle gap in the rate scale used by the per-frame surrogate problem. Therefore, minimizing this regret is consistent with (16).

This regret definition provides an analytical benchmark, but it does not imply that the joint decision should be learned as one MAB arm. Directly applying MAB to $(\mathbf{A}(t), \mathbf{M}(t))$ would treat each feasible association and scheduling pair as one arm, resulting in a combinatorial explosion in the arm space and requiring observations tied to globally coupled decisions. We therefore propose the LV-MAB framework, which factorizes the joint online learning problem into two local MAB problems coupled through association requests and scheduling feedback.

The implementation follows a sequential request and scheduling protocol. In the request step, each UA u treats its visible SA set $\mathcal{S}_u(t)$ as the local arm set, and selecting an arm means sending an association request to the selected SA. The requests generated by all UAs induce the request set observed by each SA s, which is given by

$$\mathcal{U}_s^{\text{req}}(t) = \{u \in \mathcal{U}_s(t) \mid \alpha_{us}(t) = 1\}. \tag{20}$$

In the scheduling step, each SA s treats $\mathcal{U}_s^{\text{req}}(t)$ as its local candidate arm set. Since an SA can admit multiple UAs in one frame, one scheduling action consists of selecting a subset of requesting UAs rather than a single UA. Specifically, SA s determines $\{\mu_{us}(t)\}_{u \in \mathcal{U}_s^{\text{req}}(t)}$, where $\mu_{us}(t) = 1$ means that UA u is admitted by SA s, subject to the RF chain constraint (11g). Thus, the SA side corresponds to a local combinatorial MAB problem.

We next detail the UA-side association learning and SA-side scheduling learning procedures.

1) Phase I—UA Association Learning: In this phase, each UA implements the local association MAB over its visible SA set $\mathcal{S}_u(t)$. For UA u, selecting arm $s \in \mathcal{S}_u(t)$ means sending an association request to SA s. The UA-side reward is defined according to the contribution of link (u, s) in the surrogate objective. Once $\alpha_{us}(t) = 1$, this contribution is proportional to $Q_u(t)\mu_{us}(t)R_{us}(t)\left[1 - \eta_{\text{act}}\left(1 - \mu_{us}(t-1)\right)\right]$. Since $Q_u(t)$ is common to all candidate SAs of the same UA in frame t, it does not affect the UA-side arm ranking. Moreover, the payload transmission factor $1 - \eta_{\text{act}}\left(1 - \mu_{us}(t-1)\right)$ is determined by the previous scheduling state and is available when the UA makes its request decision. Therefore, LV-MAB learns the stochastic reward $\mu_{us}(t)R_{us}(t)$ from observed rate samples and applies this factor later in the final UA-side selection index. This separation prevents the empirical mean and variance from depending on previous scheduling states rather than only on the service quality of the candidate link.

After UA u sends an association request to SA s, SA s returns admission feedback indicating whether the request is accepted. This feedback determines the scheduling variable $\mu_{us}(\tau)$. If an ACK or scheduling grant is received, then $\mu_{us}(\tau) = 1$, and UA u transmits and observes the realized service rate $R_{us}(\tau)$. If a NACK or rejection message is received, then $\mu_{us}(\tau) = 0$, and the effective service rate is zero. If no admission feedback is received within one decision frame, UA u cannot distinguish a rejection from a missing grant. In this case, UA u does not transmit in frame τ and the observation is not used to update the learned statistics.

To capture this effect, we define $\kappa_{us}(\tau) \in \{0, 1\}$ as the feedback availability indicator for link (u, s) in frame τ. For frames in which $s \in \mathcal{S}_u(\tau)$, we set $\kappa_{us}(\tau) = 1$ if either s is not requested by UA u, i.e., $\alpha_{us}(\tau) = 0$, or s is requested and the corresponding ACK/NACK feedback is received within one decision frame. Otherwise, $\kappa_{us}(\tau) = 0$. Therefore, only frames with $\alpha_{us}(\tau)\kappa_{us}(\tau) = 1$ provide valid requested rate samples for link (u, s). For these frames, the effective observed service rate is defined as

$$\tilde{R}_{us}(\tau) \triangleq \mu_{us}(\tau)R_{us}(\tau), \tag{21}$$

which equals the realized service rate when the request is accepted and zero when the request is rejected.

Considering that the effective observed rate samples $\tilde{R}_{us}(\tau)$ may have different empirical variances across candidate SAs

due to intermittent admission and non-stationary link conditions, LV-MAB adopts a variance-aware UCB rule inspired by UCB-V [24] at the UA side. In this rule, the empirical mean estimates the effective service rate of each candidate link, while the empirical variance calibrates the confidence bonus. With the same number of samples, a link with a larger empirical variance has a less reliable mean estimate and therefore receives a larger exploration bonus than a link with more stable samples.

To handle non-stationary observations, UA u maintains a link-level reset time and a sliding window for each candidate link (u,s). The reset time is initialized at the beginning of the learning process and is updated only when the statistics of the corresponding link become unreliable. Let $t^{\mathrm{UA}}_{us,\mathrm{reset}}$ denote the most recent UA-side reset frame for link (u,s). The corresponding UA-side sliding window at frame t consists of the eligible frames after this reset and within the most recent W-frame interval

$$\begin{aligned}\mathcal{W}^{\mathrm{UA}}_{us}(t) =&\{\tau : \max\{t^{\mathrm{UA}}_{us,\mathrm{reset}}, t-W\} \le \tau < t,\\ &s \in \mathcal{S}_u(\tau),\, \kappa_{us}(\tau)=1\}.\end{aligned} \tag{22}$$

Thus, an invisible frame is excluded from the UA-side statistics of link (u,s), a visible but unrequested frame is retained in the link window but contributes no rate sample, a frame with valid ACK/NACK feedback contributes one sample, and a frame with missing feedback is discarded.

The number of valid requested rate samples in $\mathcal{W}^{\mathrm{UA}}_{us}(t)$ is given by

$$N_{us}(t) = \sum_{\tau \in \mathcal{W}^{\mathrm{UA}}_{us}(t)} \alpha_{us}(\tau)\kappa_{us}(\tau). \tag{23}$$

When $N_{us}(t) > 0$, UA u computes the empirical mean and variance of the effective observed rate samples as

$$\bar{R}_{us}(t) = \frac{1}{N_{us}(t)} \sum_{\tau \in \mathcal{W}^{\mathrm{UA}}_{us}(t)} \alpha_{us}(\tau)\kappa_{us}(\tau)\tilde{R}_{us}(\tau), \tag{24}$$

and

$$\hat{\sigma}^2_{us}(t) = \frac{1}{N_{us}(t)} \sum_{\tau \in \mathcal{W}^{\mathrm{UA}}_{us}(t)} \alpha_{us}(\tau)\kappa_{us}(\tau)\left(\tilde{R}_{us}(\tau) - \bar{R}_{us}(t)\right)^2. \tag{25}$$

The variance-aware UCB estimate of the effective service rate is then given by

$$\hat{R}^{\mathrm{UCB}}_{us}(t) = \bar{R}_{us}(t) + c_1\sqrt{\frac{\hat{\sigma}^2_{us}(t)\ln|\mathcal{W}^{\mathrm{UA}}_{us}(t)|}{N_{us}(t)}} + c_2\frac{\ln|\mathcal{W}^{\mathrm{UA}}_{us}(t)|}{N_{us}(t)}, \tag{26}$$

where c_1 and c_2 are exploration coefficients.

The empirical variance is used to monitor the reliability of the current link-level window statistics. If $\hat{\sigma}^2_{us}(t) > \gamma^{\mathrm{UA}}_{\mathrm{th}}$, UA u regards the current statistics of link (u,s) as no longer representative and updates $t^{\mathrm{UA}}_{us,\mathrm{reset}} \leftarrow t$. This reset renews only the local statistics of the detected link (u,s), while the statistics of other candidate links of UA u remain unchanged.

The final UA-side selection index combines the learned rate estimate with the payload transmission factor known at the time of decision. If $N_{us}(t) = 0$, UA u sets $I_{us}(t) = +\infty$ to enforce optimistic exploration. Otherwise, the index is

$$I_{us}(t) = \hat{R}^{\mathrm{UCB}}_{us}(t)\left[1 - \eta_{\mathrm{act}}\left(1 - \mu_{us}(t-1)\right)\right]. \tag{27}$$

Algorithm 1 UA Association Learning with Variance-Aware UCB

Input: Window size W, reset threshold $\gamma^{\mathrm{UA}}_{\mathrm{th}}$, exploration coefficients c_1, c_2.

Initialize: $\mathcal{W}^{\mathrm{UA}}_{us}(0) \leftarrow \emptyset$, $N_{us}(0) \leftarrow 0$, and $t^{\mathrm{UA}}_{us,\mathrm{reset}} \leftarrow 0$ for all candidate links (u,s).
for frame $t = 1, 2, \ldots$ **do**
  for each visible SA $s \in \mathcal{S}_u(t)$ **do**
    Update $\mathcal{W}^{\mathrm{UA}}_{us}(t)$ according to (22) and compute $N_{us}(t)$ via (23).
    if $N_{us}(t) = 0$ **then**
      Set $I_{us}(t) \leftarrow +\infty$.
    else
      Compute $\bar{R}_{us}(t)$ and $\hat{\sigma}^2_{us}(t)$ via (24) and (25).
      Compute $\hat{R}^{\mathrm{UCB}}_{us}(t)$ via (26).
      Calculate $I_{us}(t)$ via (27).
  Select $s^\star$ via (28) and send an association request to SA $s^\star$ with $Q_u(t)$.
  for each $s \in \mathcal{S}_u(t) \setminus \{s^\star\}$ **do**
    Set $\alpha_{us}(t) \leftarrow 0$, $\mu_{us}(t) \leftarrow 0$, and $\kappa_{us}(t) \leftarrow 1$, and retain frame t as a visible but unrequested record.
  Set $\alpha_{us^\star}(t) \leftarrow 1$.
  if a valid **ACK** or scheduling grant is received from $s^\star$ within one decision frame **then**
    Set $\kappa_{us^\star}(t) \leftarrow 1$, $\mu_{us^\star}(t) \leftarrow 1$, and record $\tilde{R}_{us^\star}(t) = R_{us^\star}(t)$ for future window updates.
  else if a valid **NACK** is received from $s^\star$ within one decision frame **then**
    Set $\kappa_{us^\star}(t) \leftarrow 1$, $\mu_{us^\star}(t) \leftarrow 0$, and record $\tilde{R}_{us^\star}(t) = 0$ for future window updates.
  else
    Set $\kappa_{us^\star}(t) \leftarrow 0$, set $\mu_{us^\star}(t) \leftarrow 0$ in the local scheduling record, and discard frame t from the UA-side statistics of link $(u,s^\star)$.
  if $\kappa_{us^\star}(t) = 1$, $N_{us^\star}(t) > 0$, and $\hat{\sigma}^2_{us^\star}(t) > \gamma^{\mathrm{UA}}_{\mathrm{th}}$ **then**
    Set $t^{\mathrm{UA}}_{us^\star,\mathrm{reset}} \leftarrow t$, so future windows of link $(u,s^\star)$ discard records before frame t.
  Calculate $D_u(t)$ and update $Q_u(t+1)$ via (9) and (8).

UA u finally selects

$$s^\star = \arg\max_{s \in \mathcal{S}_u(t)} I_{us}(t), \tag{28}$$

where ties are broken uniformly at random. UA u then sends an association request to SA $s^\star$ with its queue backlog $Q_u(t)$ piggybacked for SA-side scheduling. The detailed UA-side procedure is summarized in Algorithm 1.

2) Phase II—SA Scheduling with CTS and Posterior Inflation: In this phase, each SA implements the local combinatorial MAB over the request set $\mathcal{U}^{\mathrm{req}}_s(t)$ defined in (20).

Since SA s must select up to $N_{\max}$ UAs from the request set under unknown link rates, the SA-side decision is a local combinatorial bandit problem. We therefore adopt CTS with a Gaussian posterior for each candidate link. The belief of link (u,s) is initialized as $\hat{\nu}_{us}(0) = \nu_{\mathrm{init}}$ and $\hat{\sigma}^2_{us}(0) = \sigma^2_{\mathrm{init}}$,

where $\hat{\nu}_{us}(t)$ and $\hat{\sigma}^2_{us}(t)$ denote the SA-side posterior mean and epistemic variance of the uplink service rate, respectively.

For each requesting UA $u \in \mathcal{U}_s^{\mathrm{req}}(t)$, SA s draws a Thompson sample

$$R_{us}^{\mathrm{TS}}(t) \sim \mathcal{N}\big(\hat{\nu}_{us}(t-1), \hat{\sigma}^2_{us}(t-1)\big), \tag{29}$$

which serves as a randomized estimate of the unknown service rate. The Gaussian posterior provides a lightweight belief representation of the unknown mean service rate and its uncertainty, rather than a physical distributional model of the instantaneous rate. The Thompson sample is then used to randomize the SA-side scheduling priority.

The sampled rate is combined with the queue backlog $Q_u(t)$ reported by UA u in its association request and the link activation cost to form the scheduling priority

$$\Psi_{us}^{\mathrm{SA}}(t) = Q_u(t) R_{us}^{\mathrm{TS}}(t) \left[1 - \eta_{\mathrm{act}} \left(1 - \mu_{us}(t-1)\right)\right]. \tag{30}$$

SA s admits the top $\min(|\mathcal{U}_s^{\mathrm{req}}(t)|, N_{\max})$ requesting UAs ranked by $\Psi_{us}^{\mathrm{SA}}(t)$ as $\mathcal{M}_s^\star(t)$, and returns ACKs or scheduling grants to admitted UAs and NACKs to the others.

An admitted request yields an SA-side rate sample only when the scheduling grant reaches the UA and the corresponding uplink transmission is established. With $\kappa_{us}(t) = 1$ indicating this event, the effective service set is defined as

$$\mathcal{U}_s^{\mathrm{srv}}(t) \triangleq \{u \in \mathcal{M}_s^\star(t) \mid \kappa_{us}(t) = 1\}, \tag{31}$$

which contains admitted UAs with established uplink transmissions before the update deadline. For $u \in \mathcal{U}_s^{\mathrm{srv}}(t)$, SA s observes the realized rate $R_{us}(t)$ from the established uplink transmission and updates the SA-side posterior as

$$\hat{\nu}_{us}(t) = \frac{\sigma_{\mathrm{o}}^2 \hat{\nu}_{us}(t-1) + \hat{\sigma}^2_{us}(t-1) R_{us}(t)}{\sigma_{\mathrm{o}}^2 + \hat{\sigma}^2_{us}(t-1)}, \tag{32}$$

$$\hat{\sigma}^2_{us}(t) = \frac{\hat{\sigma}^2_{us}(t-1)\sigma_{\mathrm{o}}^2}{\hat{\sigma}^2_{us}(t-1) + \sigma_{\mathrm{o}}^2}, \tag{33}$$

where σ_{o}^2 is the observation noise variance. For rejected UAs, the zero service outcome is induced by the scheduling decision and is not treated as a rate sample. For admitted UAs whose ACK or scheduling grant is not successfully received, the uplink transmission is not established and no rate sample is available. In both cases, the SA-side belief remains unchanged.

Similar to the UA side, SA s maintains a local reset time and a sliding window of valid rate samples for each link. Let $t_{us,\mathrm{reset}}^{\mathrm{SA}}$ denote the most recent SA-side reset frame for link (u, s). After this reset, the sliding window $\mathcal{W}_{us}^{\mathrm{SA}}(t)$ contains the latest at most W_{SA} frames τ satisfying

$$t_{us,\mathrm{reset}}^{\mathrm{SA}} \le \tau < t, \qquad \mu_{us}(\tau)\kappa_{us}(\tau) = 1. \tag{34}$$

Thus, only frames with established uplink transmission and available rate samples are included in the SA-side empirical rate statistics.

When $|\mathcal{W}_{us}^{\mathrm{SA}}(t)| > 0$, the empirical mean and variance are computed as

$$\bar{R}_{us}^{\mathrm{SA}}(t) = \frac{1}{|\mathcal{W}_{us}^{\mathrm{SA}}(t)|} \sum_{\tau \in \mathcal{W}_{us}^{\mathrm{SA}}(t)} R_{us}(\tau), \tag{35}$$

Algorithm 2 SA Scheduling with Posterior-Inflated CTS

Input: Maximum service capacity $N_{\max}$, reset threshold $\gamma_{\mathrm{th}}^{\mathrm{SA}}$, observation noise variance σ_{o}^2, window size W_{SA}.

Initialize: $\hat{\nu}_{us}(0) \leftarrow \nu_{\mathrm{init}}$, $\hat{\sigma}^2_{us}(0) \leftarrow \sigma^2_{\mathrm{init}}$, and $t_{us,\mathrm{reset}}^{\mathrm{SA}} \leftarrow 0$ for all candidate links.
for frame $t = 1, 2, \ldots$ **do**
  Identify the requesting set $\mathcal{U}_s^{\mathrm{req}}(t)$ via (20).
  for each $u \in \mathcal{U}_s^{\mathrm{req}}(t)$ **do**
    Sample $R_{us}^{\mathrm{TS}}(t) \sim \mathcal{N}(\hat{\nu}_{us}(t-1), \hat{\sigma}^2_{us}(t-1))$.
    Compute the scheduling priority $\Psi_{us}^{\mathrm{SA}}(t)$ via (30).
  Select $\mathcal{M}_s^\star(t)$ as the top-$\min(|\mathcal{U}_s^{\mathrm{req}}(t)|, N_{\max})$ UAs ranked by $\Psi_{us}^{\mathrm{SA}}(t)$.
  Send **ACK** to $u \in \mathcal{M}_s^\star(t)$ and **NACK** to $u \in \mathcal{U}_s^{\mathrm{req}}(t) \setminus \mathcal{M}_s^\star(t)$.
  Determine $\mathcal{U}_s^{\mathrm{srv}}(t) \leftarrow \{u \in \mathcal{M}_s^\star(t) \mid \kappa_{us}(t) = 1\}$.
  for each $u \in \mathcal{U}_s^{\mathrm{srv}}(t)$ **do**
    Observe the realized rate $R_{us}(t)$ and update $(\hat{\nu}_{us}(t), \hat{\sigma}^2_{us}(t))$ via (33).
    Update $\mathcal{W}_{us}^{\mathrm{SA}}(t)$ and compute $S^2_{us}(t)$ via (36).
    if $S^2_{us}(t) > \gamma_{\mathrm{th}}^{\mathrm{SA}}$ **then**
      Apply posterior inflation via (37) and set $t_{us,\mathrm{reset}}^{\mathrm{SA}} \leftarrow t$.
  Keep SA-side beliefs unchanged for rejected UAs or admitted UAs without established uplink transmissions.

and

$$S^2_{us}(t) = \frac{1}{|\mathcal{W}_{us}^{\mathrm{SA}}(t)|} \sum_{\tau \in \mathcal{W}_{us}^{\mathrm{SA}}(t)} \big(R_{us}(\tau) - \bar{R}_{us}^{\mathrm{SA}}(t)\big)^2. \tag{36}$$

The empirical variance is used to monitor the reliability of the current SA-side rate statistics. If $S^2_{us}(t) > \gamma_{\mathrm{th}}^{\mathrm{SA}}$, SA s regards the current rate statistics of link (u, s) as no longer representative and updates $t_{us,\mathrm{reset}}^{\mathrm{SA}} \leftarrow t$. For the detected link, posterior inflation is performed as

$$\hat{\nu}_{us}(t) \leftarrow R_{us}(t), \qquad \hat{\sigma}^2_{us}(t) \leftarrow \sigma^2_{\mathrm{init}}. \tag{37}$$

This operation renews the local rate statistics and restores high posterior uncertainty for rapid adaptation. The SA-side procedure is summarized in Algorithm 2.

Since $\gamma_{\mathrm{th}}^{\mathrm{UA}}$ and $\gamma_{\mathrm{th}}^{\mathrm{SA}}$ are applied to empirical rate variances, they should be expressed in consistent units and be rescaled when rate units change or when the observations are normalized. In practice, the background rate variance can be estimated during normal operation, and the thresholds can be set above this level while preserving sensitivity to abrupt changes.

C. Complexity and Overhead Analysis

We next characterize the online computation, storage, and signaling costs of LV-MAB under distributed UA-side association and SA-side scheduling.

1) Computational Complexity: For computational complexity, we consider one frame of LV-MAB. On the UA side, each UA evaluates the association index for all visible SAs in $\mathcal{S}_u(t)$ and selects the SA with the largest index. With running sums and circular buffers for the sliding-window statistics, the empirical mean, variance, and sample count can be updated incrementally. Therefore, the UA-side complexity is $\mathcal{O}(|\mathcal{S}_u(t)|)$.

On the SA side, each SA computes the CTS-based scheduling priority for every requesting UA in $\mathcal{U}_s^{\mathrm{req}}(t)$ and selects the top $N_{\max}$ UAs. With a bounded heap or partial-selection routine, the SA-side complexity is $\mathcal{O}(|\mathcal{U}_s^{\mathrm{req}}(t)| \log N_{\max} + N_{\max})$, where the second term accounts for the closed-form posterior updates of at most $N_{\max}$ admitted UAs. Since UAs and SAs operate locally within their respective phases, the per-frame decision complexity that determines online latency is bounded by $\mathcal{O}\big(\max_{u\in\mathcal{U}} |\mathcal{S}_u(t)| + \max_{s\in\mathcal{S}} \big(|\mathcal{U}_s^{\mathrm{req}}(t)| \log N_{\max} + N_{\max}\big)\big)$.

Thus, the online computation scales with local visibility and local contention, rather than with the dense global action space $|\mathcal{U}||\mathcal{S}|$.

2) Memory Footprint: The storage cost is also local. Each UA maintains scalar statistics and a sliding window of at most W retained records for each visible SA, requiring $\mathcal{O}(|\mathcal{S}_u(t)|W)$ memory. Each SA maintains posterior parameters and a valid-rate window of length at most W_{SA} for its locally relevant UA links, requiring $\mathcal{O}(|\mathcal{U}_s(t)|W_{\mathrm{SA}})$ memory. Therefore, the storage overhead is proportional to the number of locally relevant links. With 32-bit storage, the approximate memory footprints are $M_u^{\mathrm{UA}} \approx 4|\mathcal{S}_u(t)|(W+5)$ bytes and $M_s^{\mathrm{SA}} \approx 4|\mathcal{U}_s(t)|(W_{\mathrm{SA}}+5)$ bytes. Under the simulated setting with $W = W_{\mathrm{SA}} = 400$, $|\mathcal{S}_u(t)| \le 10$, and $|\mathcal{U}_s(t)| \le 50$, these values are approximately 16 KB for each UA and 79 KB for each SA.

3) Signaling Overhead: The signaling overhead is limited to local control messages. In each frame, each UA sends at most one association request carrying its queue backlog $Q_u(t)$, and each requested SA returns an ACK, a scheduling grant, or a NACK. Missing or delayed admission feedback is handled by setting the corresponding retention indicator to zero and skipping the affected learning update. Hence, no network-wide state exchange or iterative message passing is required. The total control signaling scales as $\mathcal{O}(|\mathcal{U}|)$ per frame.

Overall, LV-MAB requires only local index evaluation, local top-$N_{\max}$ selection, scalar posterior updates, and short sliding-window records. These properties make the proposed framework suitable for distributed online implementation in LEO satellite networks.

V. Theoretical Analysis

This section develops the regret analysis for LV-MAB and then connects the resulting queue-weighted service gap to queue stability through Lyapunov drift analysis.

Before presenting the regret bounds, we introduce a piecewise-stationary model for the analysis. The time horizon $[1,T]$ is partitioned into stationary epochs, and a new epoch starts when satellite motion, visibility changes, jammer retargeting, or an abrupt change in local link statistics modifies the service distribution. Let $\Gamma_{u,T}$ and $\Gamma_{s,T}$ denote the numbers of UA-side and SA-side stationary epochs over $[1,T]$, respectively. Note that this epoch partition is used only for analysis. During online operation, LV-MAB does not require these epoch counts as inputs. It updates local beliefs from ACK/NACK outcomes and locally observed rate samples retained in finite windows, and triggers resets when the empirical variance exceeds the corresponding threshold.

A. Regret Analysis

Using the policy regret in (19), we define $\mathcal{R}_{\mathrm{sys}}^{Q}(T) \triangleq \mathcal{R}_{\mathrm{sys}}^{Q,\mathrm{LV\text{-}MAB}}(T)$ for the LV-MAB policy. This quantity measures the cumulative expected gap, over T frames, between the surrogate objective value achieved by LV-MAB and that achieved by the global optimal oracle in (18). The main idea is to decompose this gap into UA-side association regret, SA-side scheduling regret, and a residual term caused by RF chain contention and queue coupling across local decisions.

Before defining the association and scheduling regret terms, we first express the effective link service under link activation cost. From the payload transmission factor in (14) and the link contribution in (16), the effective rate of link (u,s) is

$$R_{us}^{\mathrm{eff}}(t) \triangleq R_{us}(t)\left[1-\eta_{\mathrm{act}}\left(1-\mu_{us}(t-1)\right)\right]. \tag{38}$$

Moreover, since the achievable link rates are bounded, let $R_{\max}$ denote a uniform upper bound on the effective rates. A single link activation event therefore causes an effective rate reduction of at most $\omega_{\mathrm{act}} \triangleq \eta_{\mathrm{act}} R_{\max}$.

We next define the UA-side association gap using the same payload transmission factor as in (38). For the g-th stationary epoch of UA u, let $\nu_{us}^{(g)} \triangleq \mathbb{E}[\tilde{R}_{us}(t) \mid \alpha_{us}(t)\kappa_{us}(t)=1]$ denote the mean of the valid observed rate samples in (21). Let $g_u(t)$ be the UA-side epoch containing frame t, and let $s_u(t)$ be the SA requested by UA u. Applying this factor in (38) to these epoch means, we compare the best visible candidate with the requested SA and define the association gap by

$$\begin{aligned}\Delta_u^{\mathrm{assoc}}(t) \triangleq \max_{s\in\mathcal{S}_u(t)} \nu_{us}^{(g_u(t))}\left[1-\eta_{\mathrm{act}}\left(1-\mu_{us}(t-1)\right)\right] \\ - \nu_{us_u(t)}^{(g_u(t))}\left[1-\eta_{\mathrm{act}}\left(1-\mu_{us_u(t)}(t-1)\right)\right]\end{aligned} \tag{39}$$

The UA-side association regret is then

$$\mathcal{R}_u^{\mathrm{assoc}}(T) = \sum_{t=1}^{T} \mathbb{E}[\Delta_u^{\mathrm{assoc}}(t)]. \tag{40}$$

Thus, $\Delta_u^{\mathrm{assoc}}(t)$ is the per-frame association gap, while $\mathcal{R}_u^{\mathrm{assoc}}(T)$ is its cumulative expected regret.

To define the SA-side scheduling regret, we introduce a *scheduling oracle* that keeps a given association matrix fixed and optimizes only the feasible SA-side scheduling decision. Under the actual association matrix $\mathbf{A}(t)$ generated by LV-MAB, the oracle scheduling matrix is

$$\mathbf{M}^{\mathrm{sch},\star}(t) \in \arg\max_{\mathbf{M}:(\mathbf{A}(t),\mathbf{M})\in\mathcal{F}(t)} f_t(\mathbf{A}(t),\mathbf{M}). \tag{41}$$

Accordingly, let $V_t(\mathbf{A}) \triangleq \max_{\mathbf{M}:(\mathbf{A},\mathbf{M})\in\mathcal{F}(t)} f_t(\mathbf{A},\mathbf{M})$ denote the surrogate objective value attained by the scheduling oracle

under association matrix $\mathbf{A}$. With the realized association fixed, the SA-side scheduling regret of SA s is

$$\mathcal{R}_s^{\mathrm{sch},Q}(T) = \sum_{t=1}^{T} \mathbb{E}\Bigg[\sum_{u \in \mathcal{U}_s^{\mathrm{req}}(t)} Q_u(t) \mu_{us}^{\mathrm{sch},\star}(t) R_{us}^{\mathrm{eff}}(t) - \sum_{u \in \mathcal{U}_s^{\mathrm{req}}(t)} Q_u(t) \mu_{us}(t) R_{us}^{\mathrm{eff}}(t) \Bigg], \quad (42)$$

where $\mu_{us}^{\mathrm{sch},\star}(t)$ is obtained by applying the scheduling oracle to the realized LV-MAB association, and $\mu_{us}(t)$ is the LV-MAB scheduling decision.

It remains to relate the aggregate UA-side association regret to $\mathbb{E}[\sum_{t=1}^{T}(V_t(\mathbf{A}^\star(t)) - V_t(\mathbf{A}(t)))]$. Since $\sum_{u\in\mathcal{U}} \mathcal{R}_u^{\mathrm{assoc}}(T)$ sums only the UA-side rate gaps, it does not include the residual coupling caused by RF chain contention and queue interactions. With $Q_u(t) \le Q_T^{\max}$ over $[1,T]$, we define

$$\mathcal{E}_{\mathrm{coup}}(T) \triangleq \max\Bigg\{ \mathbb{E}\Bigg[\sum_{t=1}^{T} (V_t(\mathbf{A}^\star(t)) - V_t(\mathbf{A}(t)))\Bigg] - Q_T^{\max} \sum_{u\in\mathcal{U}} \mathcal{R}_u^{\mathrm{assoc}}(T), 0 \Bigg\}. \quad (43)$$

The regret decomposition is summarized in the following lemma.

Lemma 1 (Regret Decomposition). *If $Q_u(t) \le Q_T^{\max}$ over the analysis horizon, then*

$$\mathcal{R}_{\mathrm{sys}}^{Q}(T) \le Q_T^{\max} \sum_{u\in\mathcal{U}} \mathcal{R}_u^{\mathrm{assoc}}(T) + \sum_{s\in\mathcal{S}} \mathcal{R}_s^{\mathrm{sch},Q}(T) + \mathcal{E}_{\mathrm{coup}}(T). \quad (44)$$

Proof: The proof is provided in Appendix A of the supplementary material. ∎

We next bound the UA-side association term and the SA-side scheduling term.

1) UA-Side Association Regret: For UA u, define $b_{us}(t) \triangleq \hat{R}_{us}^{\mathrm{UCB}}(t) - \bar{R}_{us}(t)$ for $N_{us}(t) > 0$. To avoid mixing samples from two stationary epochs, let $\mathcal{T}_u^{\mathrm{rec}}$ denote the union of the first W frames after each UA-side epoch transition. On frames outside $\mathcal{T}_u^{\mathrm{rec}}$, the UA-side window contains only samples from the current epoch. We define the clean-window confidence event

$$\mathcal{C}_u(T) \triangleq \Big\{ |\bar{R}_{us}(t) - \nu_{us}^{(g_u(t))}| \le b_{us}(t), \ \forall t \le T,\ t \notin \mathcal{T}_u^{\mathrm{rec}}, \forall s \in \mathcal{S}_u(t) \text{ with } N_{us}(t) > 0 \Big\}. \quad (45)$$

Within each clean portion of a stationary epoch, the valid effective-rate samples $\tilde{R}_{us}(t)$ are assumed to be bounded by $R_{\max}$ and to satisfy the above Bernstein-type confidence event. The resulting bound for $\mathcal{R}_u^{\mathrm{assoc}}(T)$ is stated below.

Theorem 1 (UA-Side Association Regret Bound). *Suppose that each UA uses the sliding-window UCB-V index with window length W, and that the clean-window confidence event $\mathcal{C}_u(T)$ satisfies $\mathbb{P}(\mathcal{C}_u^c(T)) = O(T^{-2})$. Then, for any UA u, the cumulative association regret satisfies*

$$\mathcal{R}_u^{\mathrm{assoc}}(T) \le \tilde{\mathcal{O}}\Bigg(\left(\frac{T}{W} + \Gamma_{u,T}\right) \left(\sqrt{|\mathcal{S}_u|\bar{\sigma}_u^2 W} + |\mathcal{S}_u| R_{\max}\right) + \Delta_{\max}^{\mathrm{assoc}} W \Gamma_{u,T} + \omega_{\mathrm{act}} \Gamma_{u,T} \Bigg), \quad (46)$$

where $|\mathcal{S}_u| \triangleq \max_t |\mathcal{S}_u(t)|$, $\Delta_{\max}^{\mathrm{assoc}} \triangleq \max_{v\in\mathcal{U}} \max_{1\le t\le T} \Delta_v^{\mathrm{assoc}}(t)$, $\bar{\sigma}_u^2$ is a finite variance constant defined in Appendix C of the supplementary material, and $\tilde{\mathcal{O}}(\cdot)$ suppresses logarithmic factors.

Proof sketch: The detailed proof is provided in Appendix C of the supplementary material. Specifically, the proof uses the upper-bound decomposition

$$\mathcal{R}_u^{\mathrm{assoc}}(T) \le \mathcal{R}_{\mathrm{stat}} + \mathcal{R}_{\mathrm{rec}} + \mathcal{R}_{\mathrm{act}}, \quad (47)$$

where $\mathcal{R}_{\mathrm{stat}}$, $\mathcal{R}_{\mathrm{rec}}$, and $\mathcal{R}_{\mathrm{act}}$ respectively collect the contributions from suboptimal SA requests within stationary epochs, stale local samples after distribution shifts, and the effective rate reduction incurred by link activation. The first W frames after each epoch transition are charged to $\mathcal{R}_{\mathrm{rec}}$, so the stationary UCB-V argument is applied only on clean windows. Within such a window, the payload transmission factor in (27) is known at the decision time and lies in $[1-\eta_{\mathrm{act}}, 1]$. Thus the same confidence event controls the effective means after applying this factor, and the clean-window association bound in the supplementary material yields the factor $(T/W + \Gamma_{u,T})$. Boundary activation costs are bounded by $\omega_{\mathrm{act}}\Gamma_{u,T}$. Combining these bounds gives (46). □

Remark 1. *Theorem 1 shows that UA-side association can track piecewise stationary service statistics without incurring linear regret, provided that distribution shifts and link activation events occur slowly relative to the horizon. For example, choosing $W = \Theta((T/\Gamma_{u,T})^{2/3})$ yields $\mathcal{R}_u^{\mathrm{assoc}}(T) = \tilde{\mathcal{O}}(T^{2/3}\Gamma_{u,T}^{1/3} + \omega_{\mathrm{act}}\Gamma_{u,T})$. Hence the UA-side association regret is sublinear whenever $\Gamma_{u,T} = o(T)$.*

2) SA-Side Scheduling Regret: We next consider a fixed SA s. The SA-side scheduling regret in (42) is analyzed under the following observation condition.

Assumption 1 (Bounded Valid Observation Gap). *For each SA s, there exists a finite constant ℓ_s^{obs} such that, after any SA-side epoch transition at frame t_0, every affected link (u,s) that remains in the local candidate set of s has valid rate observation times $t_1, t_2, \ldots$ satisfying $t_1 \le t_0 + \ell_s^{\mathrm{obs}}$ and $t_{r+1} - t_r \le \ell_s^{\mathrm{obs}}$ while the link remains in the local candidate set.*

Assumption 1 only concerns affected links that may still be scheduled after an epoch transition. It requires such links to provide valid rate observations again within a bounded number of frames, which is a mild regularity condition when active candidate links are revisited by user requests or scheduling exploration within a bounded time.

Theorem 2 (SA-Side Scheduling Regret Bound). *Under Assumption 1, the cumulative SA-side scheduling regret of SA s over $\Gamma_{s,T}$ stationary epochs satisfies*

$$\mathcal{R}_s^{\mathrm{sch},Q}(T) \le \tilde{\mathcal{O}}\Big(Q_T^{\max} \sqrt{|\mathcal{U}_s| N_{\max} \Gamma_{s,T} T} + Q_T^{\max} \Gamma_{s,T} C_{\mathrm{reset}} \Big), \quad (48)$$

where $|\mathcal{U}_s| \triangleq \max_t |\mathcal{U}_s^{\mathrm{req}}(t)|$, and C_{reset} is a finite constant defined in Appendix B of the supplementary material.

Proof sketch: The detailed proof is provided in Appendix D of the supplementary material. Specifically, the proof uses the upper-bound decomposition

$$\mathcal{R}_s^{\mathrm{sch},Q}(T) \le \mathcal{R}_s^{\mathrm{adapt},Q}(T) + \sum_{i=1}^{\Gamma_{s,T}} \mathcal{R}_{\mathrm{stat}}^{(i),Q}(T_i), \tag{49}$$

where T_i is the length of the i-th epoch. The reset adaptation term is bounded by $Q_T^{\max}\Gamma_{s,T}C_{\mathrm{reset}}$. Within each stationary epoch, the CTS bound for the local combinatorial semi-bandit scheduling problem gives a stationary term determined by the request set size $|\mathcal{U}_s|$ and scheduling capacity $N_{\max}$. Summing the stationary terms using $\sum_i \sqrt{T_i} \le \sqrt{\Gamma_{s,T}T}$ gives (48). □

Remark 2. *Theorem 2 shows that SA-side scheduling can maintain sublinear regret under bounded valid observations and slowly varying SA-side epochs.*

3) System Regret Synthesis: We now combine Lemma 1 with the UA-side and SA-side bounds.

Theorem 3 (**System Regret Bound).** *Under the queue weight condition in Lemma 1, the confidence-event condition in Theorem 1, and Assumption 1, the cumulative expected queue-weighted system regret satisfies*

$$\begin{aligned}&\mathcal{R}_{\mathrm{sys}}^{Q}(T)\\ &\le \tilde{\mathcal{O}}\Bigg(Q_T^{\max}\left(\frac{|\mathcal{U}|T}{W}+\Gamma_\Sigma^{\mathrm{ue}}\right)\left(\sqrt{S_{\max}\bar{\sigma}^2W}+S_{\max}R_{\max}\right)\\ &+ Q_T^{\max}\sqrt{|\mathcal{S}|U_{\max}N_{\max}\Gamma_\Sigma^{\mathrm{sat}}T}\Bigg)+\mathcal{R}_{\mathrm{adapt}}^{Q}+\mathcal{E}_{\mathrm{coup}}(T),\end{aligned} \tag{50}$$

where $\Gamma_\Sigma^{\mathrm{ue}} \triangleq \sum_{u\in\mathcal{U}}\Gamma_{u,T}$, $\Gamma_\Sigma^{\mathrm{sat}} \triangleq \sum_{s\in\mathcal{S}}\Gamma_{s,T}$, $S_{\max} \triangleq \max_{u,t}|\mathcal{S}_u(t)|$, $U_{\max} \triangleq \max_{s,t}|\mathcal{U}_s^{\mathrm{req}}(t)|$, $\bar{\sigma}^2 \triangleq \max_u \bar{\sigma}_u^2$, *and* $\mathcal{R}_{\mathrm{adapt}}^{Q} \triangleq Q_T^{\max}\big[(\Delta_{\max}^{\mathrm{assoc}}W+\omega_{\mathrm{act}})\Gamma_\Sigma^{\mathrm{ue}}+C_{\mathrm{reset}}\Gamma_\Sigma^{\mathrm{sat}}\big]$.

Proof sketch: The detailed proof is provided in Appendix E. Applying Lemma 1 separates the system regret into UA-side association regret, SA-side scheduling regret, and the coupling residual. Substituting the sliding-window bound in Theorem 1 and using $|\mathcal{S}_u| \le S_{\max}$, $\bar{\sigma}_u^2 \le \bar{\sigma}^2$, and $\sum_u(T/W+\Gamma_{u,T}) = |\mathcal{U}|T/W+\Gamma_\Sigma^{\mathrm{ue}}$ gives the UA-side term and its recovery/activation contribution in $\mathcal{R}_{\mathrm{adapt}}^{Q}$. The SA-side term follows from Theorem 2 and Cauchy–Schwarz aggregation over SAs. Combining these terms with $\mathcal{E}_{\mathrm{coup}}(T)$ yields (50). □

Remark 3. *Theorem 3 provides a conditional system-regret bound. In particular, a sufficient condition for* $\mathcal{R}_{\mathrm{sys}}^{Q}(T) = o(T)$ *is* $Q_T^{\max}[(|\mathcal{U}|T/W+\Gamma_\Sigma^{\mathrm{ue}})(\sqrt{W}+1)+\sqrt{\Gamma_\Sigma^{\mathrm{sat}}T}]+\mathcal{R}_{\mathrm{adapt}}^{Q}+\mathcal{E}_{\mathrm{coup}}(T) = o(T)$. *This condition is satisfied when the local epoch counts, the finite-horizon queue-weight growth, and the coupling residual grow slowly enough relative to* T.

B. Queue Stability Analysis

Finally, we connect $\mathcal{R}_{\mathrm{sys}}^{Q}(T)$ to queue stability. To state the stability condition, we express arrivals and service in the same frame-level scale as the queue update. Let Π be the class of policies satisfying the per-frame feasibility constraints $\mathcal{F}(t)$. The corresponding frame-level stability region is

$$\begin{aligned}\Lambda \triangleq \Bigg\{&\boldsymbol{\rho}\in\mathbb{R}_+^{|\mathcal{U}|} : \exists\,\pi\in\Pi, \rho_u \le \mathbb{E}_\pi[D_u(t)\mid \mathbf{Q}(t)],\\ &\forall u\in\mathcal{U},\ \forall t\Bigg\},\end{aligned} \tag{51}$$

where the expectation is taken over the exogenous dynamics and policy randomization at frame t. Since the traffic model specifies arrivals in bits/s, the frame-level mean arrival vector is $\boldsymbol{\lambda}^{\mathrm{fr}} \triangleq \mathbb{E}[\mathbf{Y}(t)] = \tau_{\mathrm{frame}}\boldsymbol{\lambda}$. The following theorem states the stability guarantee obtained from the bound on $\mathcal{R}_{\mathrm{sys}}^{Q}(T)$.

Theorem 4 (**Queue Stability).** *Suppose that the arrival process is i.i.d. with finite second moments and that* $\boldsymbol{\lambda}^{\mathrm{fr}}$ *is strictly interior to* Λ, *i.e., there exists* $\epsilon>0$ *such that* $\boldsymbol{\lambda}^{\mathrm{fr}}+\epsilon\mathbf{1}\in\Lambda$. *Then the time average of the expected total backlog under LV-MAB satisfies*

$$\limsup_{T\to\infty}\frac{1}{T}\sum_{t=1}^{T}\sum_{u\in\mathcal{U}}\mathbb{E}[Q_u(t)] \le \frac{\Phi}{\epsilon}+\limsup_{T\to\infty}\frac{\tau_{\mathrm{frame}}\mathcal{R}_{\mathrm{sys}}^{Q}(T)}{\epsilon T}, \tag{52}$$

where $\Phi \triangleq \frac{1}{2}\sum_{u\in\mathcal{U}}\sup_t\mathbb{E}[Y_u^2(t)+D_u^2(t)]$ *is a finite quadratic moment bound on frame-level arrivals and services.*

Proof sketch: The detailed proof is provided in Appendix F of the supplementary material. Specifically, the proof combines the quadratic Lyapunov drift with the slack condition $\boldsymbol{\lambda}^{\mathrm{fr}}+\epsilon\mathbf{1}\in\Lambda$. Let $D_u^\star(t)$ denote the service under the global optimal oracle in (18). The drift separates into a negative stability term and the queue-weighted service gap $\sum_{u\in\mathcal{U}}Q_u(t)(D_u^\star(t)-D_u(t))$. Over T frames, this gap is bounded by $\tau_{\mathrm{frame}}\mathcal{R}_{\mathrm{sys}}^{Q}(T)$, since $\mathcal{R}_{\mathrm{sys}}^{Q}(T)$ is defined through the rate objective in (16). Summing the drift inequality over $t=1,\dots,T$, using the telescoping property of the Lyapunov differences, and dividing by T gives (52). □

Remark 4. *Theorem 4 shows that queue stability follows once the average regret term vanishes. In particular, if* $\mathcal{R}_{\mathrm{sys}}^{Q}(T) = o(T)$, *then LV-MAB is strongly stable and the time averaged expected backlog is bounded by* Φ/ϵ. *Thus, exploration, link activation cost, reset adaptation, and coupling effects may affect finite horizon backlog, but their average impact vanishes when the regret grows sublinearly.*

VI. Simulation Results

This section evaluates the performance of the proposed LV-MAB algorithm through simulations. To represent a dense LEO network, we simulate a mega-constellation at an altitude of $550\,\mathrm{km}$, consisting of 144 orbital planes with 88 SAs per plane, for a total of $12{,}672$ SAs. This configuration allows multiple SAs to be simultaneously visible to a typical ground UA. The ground UA distribution is modeled by a Poisson cluster process to capture spatially clustered traffic demand. Specifically, community centers are generated according to a Poisson point process within the coverage area, and UAs are uniformly distributed around these centers to form geographic traffic clusters.

Adversarial jammers are co-located with UA communities and organized into coordinated clusters. The jammers follow a spatio-temporal suppression strategy. In the spatial domain, an orthogonal targeting rule maps jammers to distinct SAs to enlarge the interference footprint. In the temporal domain, the jammer retargeting interval τ_{jam} controls how often the target set is refreshed. This setting creates time-varying interference and requires the learning algorithm to adapt online. The uplink channel is modeled using Rician fading with $K = 10\,\mathrm{dB}$, reflecting the dominant line-of-sight component in LEO links.

For numerical simulations, packet-level arrivals are generated and then converted into data volume. Let $N_u(t,m)$ denote the number of packets arriving at UA u during micro-slot m of frame t. The arrival process is modeled as $N_u(t,m) \sim \mathrm{Pois}(\lambda_u \tau_{\mathrm{slot}}/L_{\mathrm{pkt}})$, where λ_u is the average data arrival rate in bits/s and L_{pkt} is the packet size in bits. The corresponding arriving data volume is given by $Y_u(t,m) = L_{\mathrm{pkt}} N_u(t,m)$. For QoS evaluation, each UA is equipped with a finite packet buffer of size $200\,\mathrm{Mbit}$. A packet is counted as lost if it is dropped due to buffer overflow or if its queueing delay exceeds the maximum tolerable delay $D_{\max} = 2000\,\mathrm{ms}$.

Detailed simulation parameters are listed in Table I.

TABLE I
SIMULATION PARAMETERS

Notation	Value
Center frequency	$10\,\mathrm{GHz}$
Bandwidth	$400\,\mathrm{MHz}$
Noise power spectral density	$-178\,\mathrm{dBm/Hz}$
Satellite antenna array size	$(38, 38)$
Max scheduled UAs ($N_{\max}$)	5
User antenna array size	$(32, 32)$
Jammer antenna array size	$(36, 36)$
Maximum UA transmit power	$20\,\mathrm{W}$
Maximum jamming power	$20\,\mathrm{W}$
Simulation area radius	$550\,\mathrm{km}$
Rician K-factor	$10\,\mathrm{dB}$
Satellite altitude	$550\,\mathrm{km}$
Constellation scale	144 planes $\times$ 88 SAs
Total satellites	$12,672$
User density	$3.68 \times 10^{-5}\,\mathrm{km}^{-2}$
Jammer density	$5 \times 10^{-4}\,\mathrm{km}^{-2}$
Micro-slot duration (τ_{slot})	$2\,\mathrm{ms}$
Decision interval (τ_{frame})	$100\,\mathrm{ms}$ ($50\ \tau_{\mathrm{slot}}$)
Link activation cost (C_{act})	$2\ \tau_{\mathrm{slot}}$ ($4\,\mathrm{ms}$)
Traffic arrival process	Poisson process (λ_u)
Packet size (L_{pkt})	$8\,\mathrm{kbit}$
Finite buffer size	$200\,\mathrm{Mbit}$
Delay deadline ($D_{\max}$)	$2000\,\mathrm{ms}$

A. Benchmark Schemes

We compare the proposed framework with five benchmark schemes, including stationary, non-stationary, and adversarial learning methods, as well as heuristic association rules.

- *UCB*: A classic stationary algorithm utilizing full history, serving as a baseline to demonstrate the necessity of forgetting mechanisms.
- *SW-UCB*: A sliding-window approach for non-stationary environments. It discards stale data but lacks variance-aware detection.
- *EXP3*: A prominent adversarial bandit algorithm designed for non-stochastic rewards, serving as a robust benchmark against intelligent jamming.
- *Nearest*: A geometry-based heuristic that associates with the closest SA, ignoring channel quality and jamming status.
- *Random*: A lower-bound baseline where UAs randomly select visible SAs without intelligent decision-making.

To ensure a fair and reproducible comparison, all schemes are evaluated under the same constellation geometry, visibility constraints, traffic arrival process, jammer process, link activation cost, RF chain limit, and decision-frame structure. The baselines differ from LV-MAB mainly in the UA-side association rule and in whether the SA side uses CTS-based rate estimation. UCB, SW-UCB, EXP3, and LV-MAB all scan the locally visible SAs at the UA side, which gives the same association complexity $\mathcal{O}(|\mathcal{S}_u(t)|)$. Nearest and Random require no learned state. For tunable learning baselines, including UCB, SW-UCB, and EXP3, the parameters specific to each algorithm are selected from standard ranges, and the best-performing settings are reported.

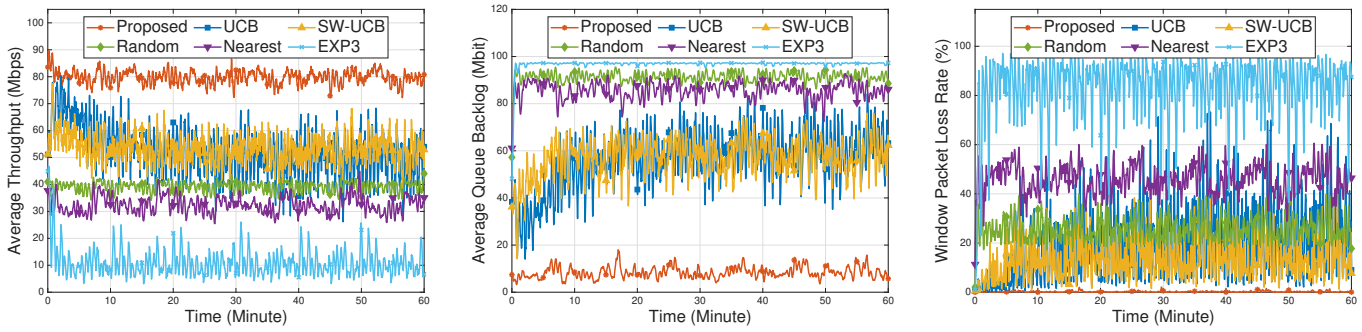
After association requests are generated, all baseline schemes use the same backlog-based scheduler at each SA. Since these baselines do not use CTS-based rate estimates, the scheduler selects the $\min(|\mathcal{U}_s^{\mathrm{req}}(t)|, N_{\max})$ requesting UAs with the largest backlogs $Q_u(t)$. This common SA-side rule keeps the scheduling resources and queue information identical across baselines. The remaining performance gap mainly reflects the benefit of LV-MAB's variance-triggered association adaptation and CTS-based rate-aware scheduling. The key algorithmic parameters of LV-MAB and the tuning settings for the learning baselines are summarized in Table II.

TABLE II
ALGORITHMIC PARAMETERS AND BASELINE TUNING

Scheme	Parameter	Value / Tuning Range
LV-MAB	Window size (W, W_{SA})	$(400, 400)$
LV-MAB	Reset thresholds $(\gamma_{\mathrm{th}}^{\mathrm{UA}}, \gamma_{\mathrm{th}}^{\mathrm{SA}})$	$(10, 10)$ $(\mathrm{Mbps})^2$
LV-MAB	Exploration coefficient (c_1, c_2)	$(0.1, 1)$
LV-MAB	Prior mean ν_{init}	$R_{\max}/2$
LV-MAB	Prior variance σ_{init}^2	$(R_{\max}/8)^2$
LV-MAB	Observation noise variance σ_o^2	1 $(\mathrm{Mbps})^2$
UCB	Exploration coefficient	1 / $\{0.7, 1, 2\}$
SW-UCB	Exploration coefficient	1 / $\{0.7, 1, 2\}$
SW-UCB	Window size	400 / $\{400, 600, 800\}$
EXP3	Exploration/learning parameter	0.1 / $\{0.05, 0.1, 0.2\}$

B. Temporal Dynamics

Fig. 2 characterizes LV-MAB's temporal response under dynamic jamming. Specifically, Fig. 2(a) shows that LV-MAB maintains an average throughput of about $80\,\mathrm{Mbps}$, while Nearest and Random remain below $40\,\mathrm{Mbps}$. UCB experiences gradual throughput degradation because its full-history estimates retain stale observations, and EXP3 shows stronger oscillations due to adversarial exploration. This throughput gain is reflected in the queue dynamics in Fig. 2(b). Nearest and Random quickly build up a large backlog of about



(a) Average throughput evolution. (b) Average queue backlog evolution. (c) Packet loss rate evolution.

Fig. 2. Temporal evolution of network performance metrics over a 60-minute window under dynamic target-hopping jamming. (a) Average throughput, illustrating the proposed framework's resilience to non-stationary interference compared to baselines. (b) Queue backlog dynamics, showing backlog control under Lyapunov-guided scheduling. (c) Packet loss rate, demonstrating reliable service continuity under non-stationary interference.

97.25 Mbit; this value is not the buffer limit, but a timeout-regulated level, since packets that wait too long are discarded once their delay exceeds the expiration threshold. In contrast, LV-MAB keeps the average backlog around 6.36 Mbit, indicating that it can serve packets before timeout occurs. Consistently, Fig. 2(c) shows that LV-MAB achieves a lower packet loss rate, whereas the losses of the heuristic schemes mainly come from packet expiration caused by excessive queueing delay.

To examine the short time scale adaptation hidden by the long horizon average, we further zoom into the initial 30 s interval, where the algorithms experience cold start and a jammer retargeting event.

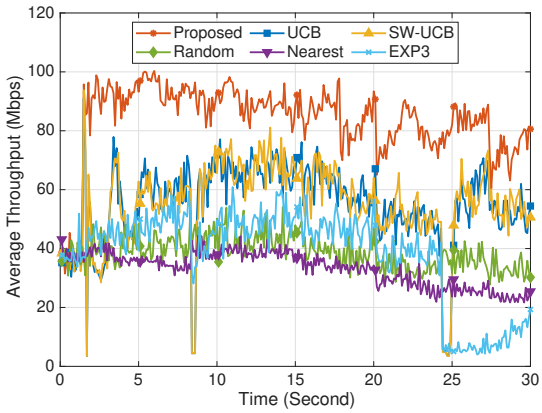


Fig. 3. Short time scale throughput response during the initial 30 s interval.

Fig. 3 characterizes the short time scale throughput response within the initial 30 s interval. In this trace, LV-MAB reaches a high-throughput regime of approximately 95 Mbps within about 2 s and recovers rapidly after the jammer retargeting event at $t \approx 24$ s. This behavior is consistent with the variance-aware exploration and reset design. During cold start, the empirical variance $\hat{\sigma}^2_{us}$ helps calibrate the uncertainty of local rate estimates. After the jamming pattern changes, the variance increase on the affected link triggers link-level belief renewal and guides the UA toward a less jammed link. Compared with UCB and SW-UCB, which show longer throughput degradation in this trace due to stale statistics, LV-MAB adapts more promptly. This short-window trace therefore illustrates how variance-triggered reset mitigates stale-statistic effects more effectively than passive window sliding in highly dynamic LEO scenarios.

C. Scalability Analysis

We next evaluate whether LV-MAB remains stable as service demand and access contention increase.

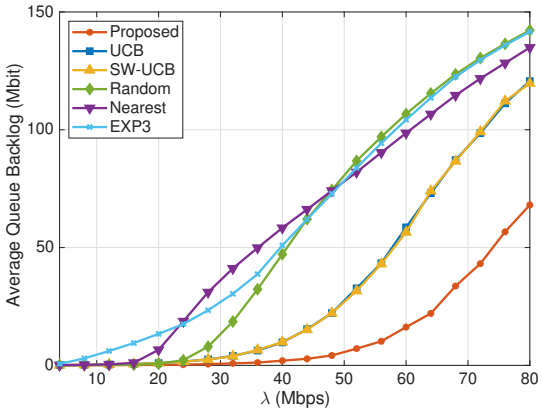


Fig. 4. Average queue backlog versus traffic arrival rate λ.

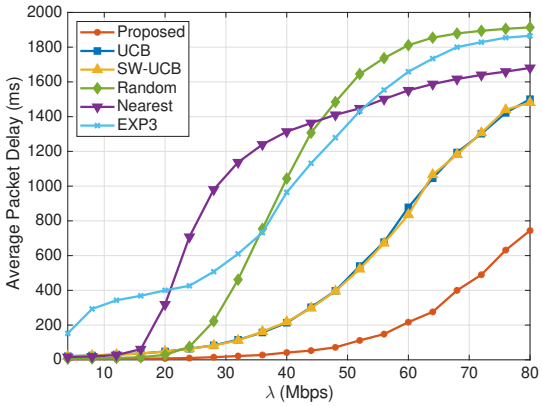


Fig. 5. Average delay of successfully delivered packets versus traffic arrival rate λ.

Figs. 4 and 5 show the impact of increasing traffic load. As λ grows, LV-MAB exhibits the slowest increase in both queue backlog and packet delay, indicating that it can support heavier traffic while maintaining stable service. This behavior follows from the coupling between Lyapunov queue weighting and CTS-based rate learning. The queue weight raises the priority of UAs with accumulated backlogs, while rate learning helps SAs avoid repeatedly serving links with poor service. In contrast, EXP3, Nearest, and Random enter congestion earlier. UCB and SW-UCB also deteriorate under high traffic load because their backlog-based scheduling decisions are not supported by sufficiently responsive rate estimation. These results show that LV-MAB provides a larger stable operating range and lower packet delay as the offered load increases.

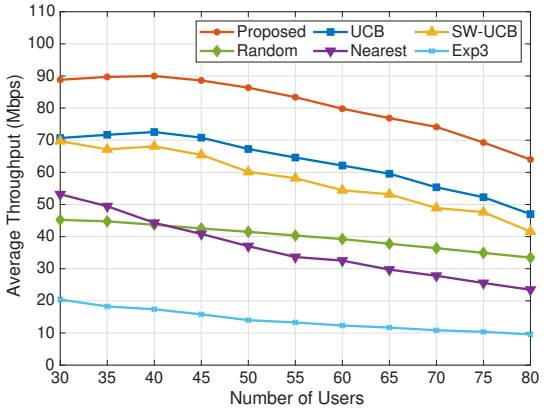


Fig. 6. Average throughput versus total number of UAs ($N = 30$ to 80).

Fig. 6 shows the throughput under increasing user density. As the number of UAs increases from 30 to 80, all schemes lose throughput because more users compete for the same RF chains and visible SAs. LV-MAB remains the best performing scheme and decreases more slowly, staying around $64\,\mathrm{Mbps}$ at $N = 80$. The Nearest strategy degrades most sharply and eventually falls below Random, because many UAs attach to nearby SAs and create local congestion. These results indicate that LV-MAB scales better with user density by balancing link quality and resource contention.

D. Fairness and Service Accessibility

To evaluate user fairness, we adopt Jain's Fairness Index $F = (\sum_{u\in\mathcal{U}} \bar{R}_u)^2/(|\mathcal{U}| \sum_{u\in\mathcal{U}} \bar{R}_u^2)$, where $\bar{R}_u$ denotes the average served rate of UA u. As shown in Fig. 7, LV-MAB keeps F close to 0.98, indicating that its throughput gain is not obtained by starving a subset of UAs. The queue weight in the scheduling utility raises the priority of users with accumulated backlogs, which helps balance service across UAs. Random also obtains a high fairness index, but this comes at the expense of much lower aggregate throughput. Nearest has a lower fairness index of about 0.90, because proximity-based association concentrates requests around nearby SAs and leaves some users underserved.

Considering that fairness in delivered rate does not fully describe whether users can access service when the network is congested or jammed, Fig. 8 reports the rejected-UA ratio

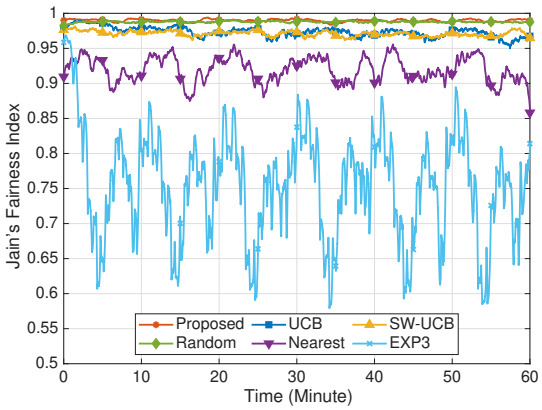


Fig. 7. Temporal evolution of Jain's Fairness Index.

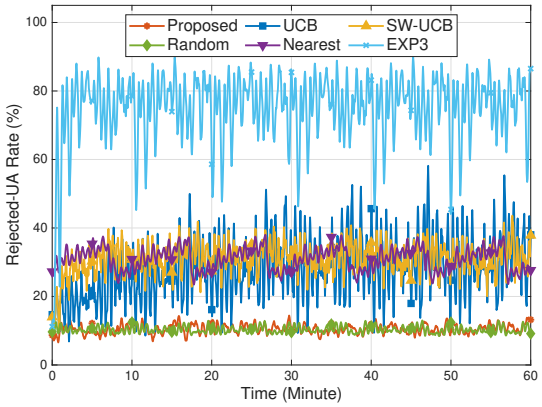


Fig. 8. Temporal evolution of the rejected-UA ratio.

under dynamic jamming. LV-MAB maintains a lower rejection ratio by combining backlog pressure with variance-triggered resets. Queued UAs receive higher scheduling priority, and users are less likely to remain attached to degraded links after jamming changes the affected link-level rate statistics. In contrast, the benchmark schemes exhibit more rejection spikes when requests concentrate on overloaded SAs or when stale statistics keep users attached to poor links. Together, Figs. 7 and 8 show that LV-MAB improves both delivered-rate fairness and service accessibility.

E. Robustness Analysis

Figs. 9 and 10 evaluate two jamming stress factors. The first is temporal variation, controlled by the jammer retargeting interval τ_{jam}. A smaller τ_{jam} means that the jammer changes its target more frequently. As shown in Fig. 9, the learning baselines degrade when τ_{jam} is small because their rate estimates lag behind the changing interference pattern. LV-MAB maintains higher throughput by using variance-triggered resets to renew stale statistics on affected links. The second factor is spatial jamming intensity. In Fig. 10, throughput decreases as the number of jammers per cluster increases, since more SAs are exposed to coordinated interference. LV-MAB shows the slowest degradation and still maintains about $80.6\,\mathrm{Mbps}$ under 12 jammers per cluster.

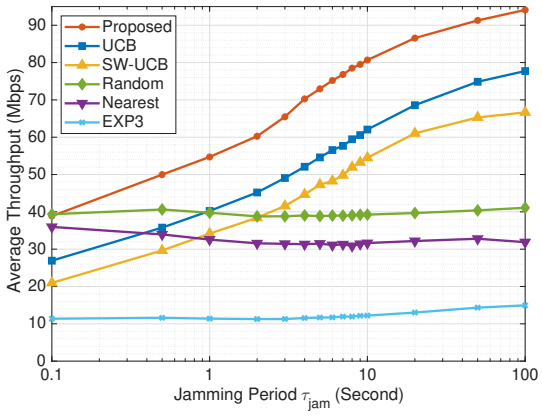


Fig. 9. Average throughput versus jammer retargeting interval τ_{jam}.

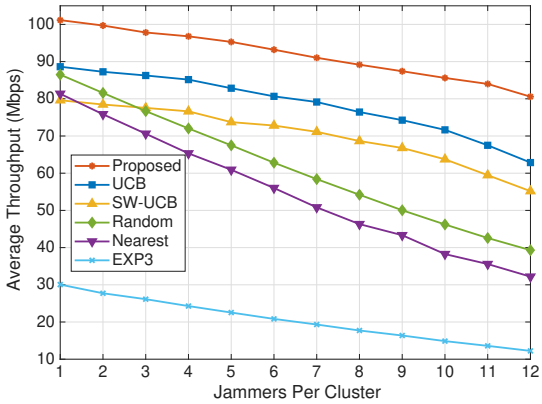


Fig. 10. Average throughput versus the number of jammers per cluster.

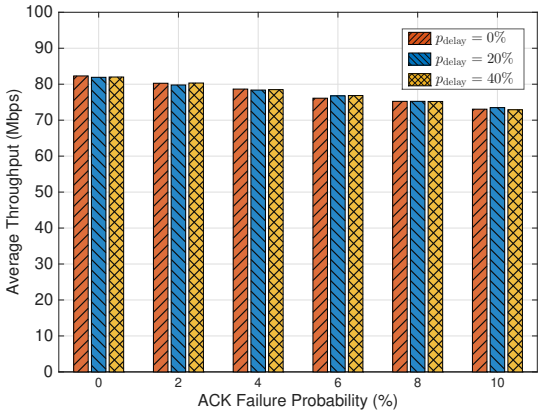


Fig. 11. Impact of ACK/NACK feedback failure and delayed feedback on the average throughput.

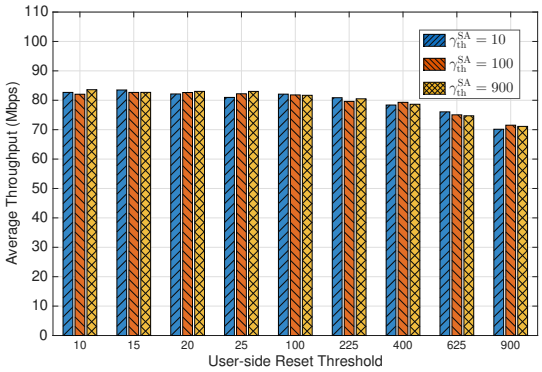


Fig. 12. Sensitivity of LV-MAB to UA-side and SA-side reset thresholds.

Considering that ACK/NACK messages may be unavailable or delayed in practical control links, we also evaluate LV-MAB under imperfect feedback. The ACK/NACK failure probability denotes the probability that the feedback message is unavailable, and the delay probability p_{delay} denotes the probability that feedback is delayed within one frame. Accordingly, delayed or unavailable feedback is discarded following the learning procedure described earlier. As shown in Fig. 11, increasing the ACK/NACK failure probability causes only a moderate throughput reduction. For a fixed failure probability, changing p_{delay} leads to minor variations, indicating that occasional delayed feedback has limited impact on LV-MAB.

F. Sensitivity and Ablation Study

Finally, we evaluate how the reset thresholds affect throughput and separate the contribution of variance-triggered reset and CTS-based scheduling through an ablation study.

Fig. 12 evaluates the sensitivity of LV-MAB to the UA-side reset threshold $\gamma_{\text{th}}^{\text{UA}}$ and the SA-side reset threshold $\gamma_{\text{th}}^{\text{SA}}$. The average throughput remains nearly stable when $\gamma_{\text{th}}^{\text{UA}}$ lies in a small-to-moderate range, which indicates that LV-MAB does not rely on fine threshold tuning. An excessively large $\gamma_{\text{th}}^{\text{UA}}$ leads to visible throughput degradation, because delayed link-level reset causes stale association statistics on degraded links to persist after the jamming pattern changes. By comparison, varying $\gamma_{\text{th}}^{\text{SA}}$ introduces only small fluctuations, since the SA-side reset mainly refines link-level local rate beliefs for scheduling rather than changing the primary association decision.

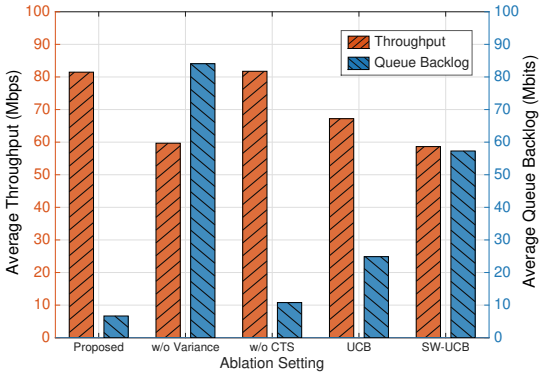


Fig. 13. Ablation study of variance-triggered reset and CTS-based scheduling.

Fig. 13 isolates the contribution of the main algorithmic mechanisms. Removing the variance-triggered reset causes the average throughput to drop from approximately $81.4\,\text{Mbps}$ to $59.6\,\text{Mbps}$, while the average queue backlog increases to around $84.1\,\text{Mbit}$. This result shows that renewing stale local statistics is critical under non-stationary jamming. The CTS-based scheduling mechanism provides a smaller but still useful gain. Compared with the complete LV-MAB framework, the

variant without CTS maintains a similar throughput level but has a slightly higher queue backlog, suggesting that CTS-based rate estimation mainly improves user scheduling and queue clearance. Overall, these results indicate that the main robustness of LV-MAB comes from variance-triggered reset, while CTS-based scheduling further improves queue clearance under RF chain constraints.

VII. Conclusion

This paper investigated resilient user association and satellite-side scheduling for LEO access under adversarial jamming, link activation cost, finite RF chains, and long-term queue stability requirements. We proposed LV-MAB, a decentralized online learning framework that converts queue stability into a per-frame queue-weighted surrogate objective and factorizes the resulting decision into UA-side association and SA-side scheduling. By combining variance-aware link-level adaptation with CTS-based scheduling, LV-MAB learns from local observations while tracking non-stationary rate statistics. We established dynamic regret bounds, proved queue stability, and demonstrated throughput, backlog, fairness, scalability, and robustness gains under dynamic jamming. A possible future direction is cooperative multi-hop LEO access, where relay and path selection as well as inter-hop queue coupling introduce additional design challenges.

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